DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 3: Yield Curves

Term Structure · Spot & Forward Rates · Bootstrapping

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Today's Agenda

1. Why one interest rate is not enough
2. Spot rates, par yields, forward rates
3. Bootstrapping: peeling the onion
4. What the curve's *shape* tells us
5. Straight into the Lab: code a bootstrapper

Learning Objectives

- Explain why bond markets need a *curve*, not a single rate.
- Distinguish spot, par, and forward rates – and convert between them.
- Explain, step by step, how bootstrapping recovers the zero curve.
- Interpret curve shape via the three term-structure theories.

A Menu of Rates, Not a Single Price

The Fixed-Deposit Counter

A bank doesn't quote one rate for all deposits. It quotes a *menu*: 6.5% for 1 year, 7.0% for 3 years, 7.2% for 5 years... The **yield curve** is exactly this menu, made precise, for the safest borrower in the economy: the government.

Term Structure: Definition

Definition

The **term structure of interest rates** is the relationship between yield on default-free debt and time to maturity. Plotted, $y(t)$ vs. t, this is the **yield curve**.

Three Related but Different Curves

- **Spot rate** $z(t)$: rate on money invested today, with a single lump-sum repayment at t. $=$ YTM of a hypothetical zero-coupon bond.
- **Par yield**: coupon rate making a bond of maturity t trade exactly at par today. What we typically *observe*.
- **Forward rate** $f(t_1, t_2)$: rate, agreed *today*, applying between two *future* dates. *Implied*, not directly observed.

Spot vs. Forward: Booking a Cab

A Cab Ride Now vs. a Cab Ride Booked in Advance

A spot rate: the fare for a ride starting *right now*. A forward rate: booking, today, the fare for a ride that starts *later* and ends even later still. You haven't taken the ride yet – but the price is locked in today, consistent with today's fares.

Discount Factors

Definition: Discount Factor

$$d(t) = \frac{1}{(1 + z(t))^t} \qquad \text{or} \qquad d(t) = e^{-z(t) \cdot t}$$

"How many rupees today for Re. 1 at time t?"

Know $d(t)$ for every $t \Rightarrow$ price *any* default-free cash-flow stream: $PV = \sum_t CF_t \cdot d(t)$.

Forward Rates from Spot Rates

No-arbitrage consistency: invest t_2 years directly $=$ invest t_1 years then reinvest at the forward rate.

$$(1 + z(t_2))^{t_2} = (1 + z(t_1))^{t_1} \left(1 + f(t_1, t_2)\right)^{t_2 - t_1}$$

$$f(t_1, t_2) = \left[\frac{(1 + z(t_2))^{t_2}}{(1 + z(t_1))^{t_1}}\right]^{\frac{1}{t_2 - t_1}} - 1$$

Example: Computing a Forward Rate

Given

$z(1) = 6\%$, $z(2) = 6.5\%$. Find $f(1,2)$.

$$f(1,2) = \frac{(1.065)^2}{(1.06)^1} - 1 = \frac{1.134225}{1.06} - 1 \approx \mathbf{7.0\%}$$

Higher than both spot rates – always true when the spot curve is upward-sloping.

Bootstrapping: Peeling the Onion

Peeling an Onion, One Layer at a Time

A 3-year coupon bond bundles *three* cash flows – each deserving its own rate. One bond price can't pin down three unknowns. But if you already know the 1-yr and 2-yr zero rates, the 3-yr bond's price has just *one* new unknown: the 3-yr rate. Solve shortest-to-longest, one new rate at a time.

The Bootstrapping Algorithm

1. **Shortest maturity**: solve $z(t_1)$ directly (single cash flow, or direct T-Bill yield).
2. **Step** k: $P_k = \sum_{i=1}^{k-1} \frac{C_k}{(1+z(t_i))^{t_i}} + \frac{C_k + F}{(1+z(t_k))^{t_k}}$ – every $z(t_i)$, $i < k$ already known
   $\Rightarrow$ **one** equation, **one** unknown: $z(t_k)$.
3. **Repeat** to the longest maturity; **interpolate** between knot points for a continuous $z(t)$.

Worked Example: Two-Step Bootstrap

Bond A – 1yr, coupon 5%, price 99.50, Face Value = 100

$$99.50 = \frac{105}{1 + z(1)} \Rightarrow z(1) \approx \mathbf{5.528\%}$$

Bond B – 2yr, coupon 5.5%, price 99.80, Face Value = 100

$$99.80 = \frac{5.5}{1.05528} + \frac{105.5}{(1 + z(2))^2}$$

$$94.588 = \frac{105.5}{(1 + z(2))^2} \Rightarrow z(2) \approx \mathbf{5.606\%}$$

Exactly what you'll automate, across many more maturities, in the Lab.

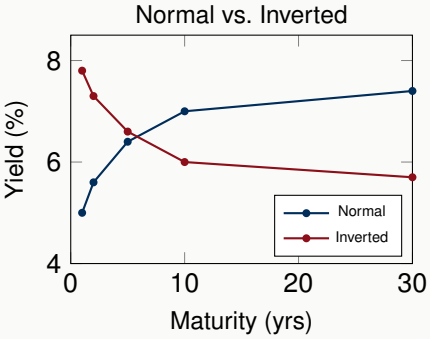
Why Not Just Plot YTM?

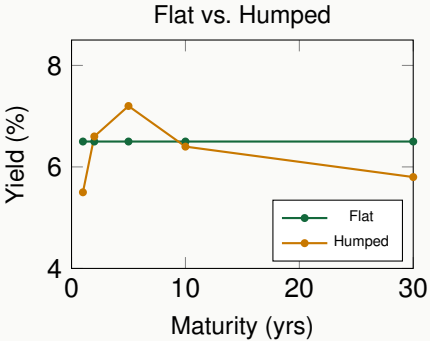
Remark

Plotting each bond's YTM against maturity is tempting – and gives a quick *par yield curve*. But YTM blends every cash flow of *one specific bond* at the *same* rate – only an approximation once the true curve isn't flat.

The zero curve $z(t)$ is the internally consistent object that correctly prices *every* instrument simultaneously.

The Shape of the Curve





Three Theories of the Term Structure

- **Expectations Hypothesis**: long rates $\approx$ average of expected future short rates. Upward slope $\Rightarrow$ rates expected to *rise*; inversion $\Rightarrow$ rates expected to *fall* (recession signal).
- **Liquidity Preference**: investors demand a term premium for locking up money longer $\Rightarrow$ curve is *normally* upward-sloping even absent any expectation of rate changes.
- **Market Segmentation**: different investors have structural preferences for specific maturity buckets $\Rightarrow$ local supply/demand can create humps and kinks.

The Curve as an Economic Thermometer

A Quick, Imperfect Read on Economic Health

A doctor reads temperature as a fast signal of health. Economists read the curve's *slope* the same way: steep-normal $\Rightarrow$ confidence in growth; flattening/inverting $\Rightarrow$ market pricing in a slowdown. No single indicator is perfect – but few are watched as closely.

Key Takeaways

Key Takeaway

- A single rate cannot describe a bond market – the **yield curve** $z(t)$ is a full function of maturity.
- **Spot**: one future cash flow. **Par**: observed coupon-bond yields. **Forward**: implied future-period rates.
- **Bootstrapping**: solve one new maturity's rate at a time, using only previously solved shorter rates.
- Curve *shape* reflects rate expectations, term premia, and maturity-specific segmentation.

Check Your Understanding

Before the Lab

1. Given $z(1) = 5\%$, $z(2) = 5.4\%$, $z(3) = 5.7\%$: compute $f(2, 3)$.
2. A 2-yr, 8%-coupon bond trades at par. What does this tell you about $z(2)$ immediately – no bootstrapping algebra?
3. Sketch the curve you'd expect when the central bank has been hiking short rates, but markets expect a recession and cuts within 2 years.

Straight Into the Lab

You now have every piece:

bond pricing (L2) + spot/par/forward definitions + the bootstrap recursion (L3)

Let's code it.

Build a bootstrapper against real Treasury par-yield data, interpolate a continuous curve, and validate it by pricing an off-benchmark bond.